ants were beginning to attack the picnic in droves. Price would have to move fast.

He didn't. Others beat him to the answers. He got there third, after Novikov and after Israel, but he got there more firmly, more completely, with deeper insight.

Price's insight was immortalized by Jack Smith, a humorous columnist for the *Los Angeles Times*. In the 27 August 1970 issue of the *Times,* Smith described a visit the previous day to Caltech: "After luncheon at the Faculty Club I walked alone around the campus. I could feel the deep thought in the air. Even in summer it stirs the olive trees. I looked in a window. A blackboard was covered with equations, thick as leaves on a walk, and three sentences in English: *Price's Theorem: Whatever can be radiated is radiated Schutz's Obseroation: Whatever is radiated can be radiated Things can be radiated if and only if they are radiated.* I walked on, wondering how it will affect Caltech this fall when they let girls in as freshmen for the first time. I don't think they'll do the place a bit of harm ... I have a hunch they'll radiate."

This quote requires some explanation. "Schutz's observation" was facetious, but Price's theorem, "Whatever can be radiated is radiated," was a serious confirmation of a 1969 speculation by Roger Penrose.

Price's theorem is illustrated by the implosion of a mountain-endowed star. Figure 7.4 depicts the implosion. The left half of this figure is a spacetime diagram of the type introduced in Figure 6.7 of Chapter 6; the right side is a sequence of snapshots of the star's and horizon's shape as time passes, with the earliest times at the bottom and the latest at the top.

As the star implodes (bottom two snapshots in Figure 7.4), its mountain grows larger, producing a growing, mountain-shaped distortion in the star's spacetime curvature. Then, as the star sinks inside its critical circumference and creates a black hole horizon around itself (middle snapshot), the distorted spacetime curvature deforms the horizon, giving it a mountain-like protrusion. The horizon's protrusion, however, cannot live long. The stellar mountain that generated it is now inside the hole, so the horizon can no longer feel the mountain's influence. The horizon is no longer being forced, by the mountain, to keep its protrusion. The horizon ejects the protrusion in the only way it can: It converts the protrusion into ripples of spacetime curvature (gravitational waves—Chapter 10) that propagate away in all directions (top two snapshots). Some of the ripples go down the hole, others